

Exercise 5: Stochastic processes & Comparison inequalities

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Problem 1 (Brownian motion):

1. Let B_t be standard Brownian motion, i.e. $B_t \sim N(0, t)$ has density.

$$f_{B_t}(x) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{x^2}{2t}\right).$$

Show that

$$P(B_t \geq a) = 1 - \Phi\left(\frac{a}{\sqrt{t}}\right),$$

where

$$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du$$

is the standard normal CDF.

Hint: $P(B_t \geq a) = \int_{a/\sqrt{t}}^{\infty} \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{u^2}{2}\right) (\sqrt{t} du)$

2. Using your previous answer, show that

$$\mathbb{E} \sup_{t \leq t_0} B_t = \sqrt{\frac{2t_0}{\pi}}, \quad t_0 \geq 0.$$

Problem 2 (Outer-Product Distance Inequality):

Let $u, v, w, z \in \mathbb{R}^n$ be unit vectors, i.e. $\|u\|_2 = \|v\|_2 = \|w\|_2 = \|z\|_2 = 1$. Prove that

$$\|uv^\top - wz^\top\|_F^2 \leq \|u - w\|_2^2 + \|v - z\|_2^2.$$

1. Show that for any $a \in \mathbb{R}^n, b \in \mathbb{R}^m$,

$$\|a b^\top\|_F = \|a\|_2 \|b\|_2.$$

Deduce that

$$\|uv^\top - wz^\top\|_F^2 = 2 - 2(u^\top w)(v^\top z).$$

2. Verify that

$$\|u - w\|_2^2 = 2 - 2u^\top w, \quad \|v - z\|_2^2 = 2 - 2v^\top z,$$

and hence

$$\|u - w\|_2^2 + \|v - z\|_2^2 = 4 - 2(u^\top w + v^\top z).$$

3. Show that

$$\|uv^\top - wz^\top\|_F^2 = (\|u - w\|_2^2 + \|v - z\|_2^2) = 2[1 - u^\top w][1 - v^\top z].$$

4. Conclude the desired inequality by observing that each factor $1 - u^\top w$ and $1 - v^\top z$ is nonnegative.

Problem 3 (Gaussian Covariance Interpolation):

Define a Gaussian random vector $Z(u)$ in $\mathbb{R}^n$ that continuously interpolates between $Z(0) = Y$ and $Z(1) = X$:

$$Z(u) := \sqrt{u} X + \sqrt{1-u} Y, \quad u \in [0, 1].$$

Prove that the covariance matrix of $Z(u)$ interpolates linearly between the covariance matrices of Y and X , namely

$$\Sigma(Z(u)) = u \Sigma(X) + (1-u) \Sigma(Y).$$